Part-Aware Multimodal Temporal Modelling with Head-Token Mixing Attention for Indian Sign Language Recognition

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Abstract: Indian Sign Language (ISL) is used by millions of deaf people in India, yet automatic ISL recognition has received far less attention than American or German sign language recognition. This paper presents a landmark-based model for isolated ISL recognition that combines part-aware multi-scale temporal convolution, gated multimodal fusion and Head-Token Mixing Attention (HTMA). Each video is uniformly sampled into 64 frames. For every frame, MediaPipe landmarks of both hands, the body pose and selected face points form a 534-dimensional landmark vector, which is complemented by 469 engineered geometry features (hand-centric coordinates, bone lengths, joint angles, spatial relationships, hand velocity and acceleration) and a four-dimensional availability indicator for missing detections. Separate encoders built from residual depthwise temporal Conv1D blocks process the left-hand, right-hand, pose and face streams; a learned gate weights these streams and the geometry stream before they are projected to a 320-dimensional sequence. Four HTMA blocks model the sequence, and a CLS-token representation combined with learned attention pooling feeds a 262-class classifier. On the locked test set of the official INCLUDE partition (816 videos, 262 classes), the model achieves 98.04% Top-1 accuracy, 99.51% Top-5 accuracy, 98.09% macro-F1 and 98.19% balanced accuracy. In the accompanying application, the recognised sign is displayed as text and spoken through the browser Web Speech API. We relate these results to previously reported INCLUDE results and discuss the limitations of the evaluation.

Key Words: Indian Sign Language, INCLUDE, isolated sign language recognition, MediaPipe landmarks, part-aware temporal convolution, gated multimodal fusion, attention, text-to-speech
Category: I.2.10, I.4.8, I.5.4
DOI: To be assigned by J.UCS

1 Introduction

Sign languages are complete natural languages with their own grammar, syntax and lexicon, and they are the first language of many deaf people. Indian Sign Language (ISL) is used by more than five million deaf people in India and differs from Western sign languages in several ways, including a highly iconic lexicon, pervasive

compound signs and a distinctive use of the signing space; it is also largely two-handed [Sridhar et al. 2020]. Because few hearing people know ISL, everyday interactions in schools, hospitals, workplaces and public offices frequently depend on the availability of a human interpreter. Automatic sign language recognition (SLR), which maps a video of a sign to its meaning, is one step towards reducing this barrier.

SLR is commonly divided into isolated recognition, in which each video contains a single sign, and continuous recognition, in which a video contains a sequence of signs that must be segmented and recognised [Sridhar et al. 2020, Camgöz et al. 2020]. Even isolated recognition is difficult. Signs are short, typically two to four seconds; many signs differ only in subtle finger, hand or arm movements, so inter-class variance is low; and individual signers perform the same sign differently, so intra-class variance is high [Sridhar et al. 2020]. Accuracy also falls as the vocabulary grows: on WLASL, the same models score far lower on 2,000 glosses than on 100 [Li et al. 2020].

Two families of input representation dominate the literature. Appearance-based methods learn directly from RGB frames with 2D or 3D convolutional networks such as I3D [Li et al. 2020, Carreira and Zisserman 2017]; they capture dense visual detail but are computationally demanding. Keypoint-based methods first estimate body and hand landmarks with an off-the-shelf estimator such as OpenPose [Cao et al. 2017] or MediaPipe [Lugaresi et al. 2019], and then classify the landmark sequence. Keypoints are compact and inexpensive to process [Boháček and Hrúz 2022], but the design choices around them vary considerably. Several systems restrict the representation to particular body regions: the INCLUDE baselines discard facial keypoints [Sridhar et al. 2020], and SignBERT models hand pose [Hu et al. 2021]. Multiple parts are often concatenated into one vector per frame [Boháček and Hrúz 2022, Li et al. 2020]. Missing detections are handled by zero-filling without an explicit indicator [Boháček and Hrúz 2022] or by optional imputation [Sridhar et al. 2020], and temporal information is aggregated by average pooling [Li et al. 2020], a single class query [Boháček and Hrúz 2022] or attention-weighted summation [Hu et al. 2021].

ISL has received comparatively little attention. The INCLUDE dataset [Sridhar et al. 2020] provided the first large, publicly available ISL benchmark, and its best reported model reached 85.6% accuracy over 263 signs. Most Transformer-based SLR systems, in contrast, have been developed for American, Argentinian, Chinese and German sign languages [Boháček and Hrúz 2022, Hu et al. 2021, Camgöz et al. 2020].

This paper describes a landmark-based model for isolated ISL recognition, implemented as IncludePartAwareConvHTMAV2 and referred to here as a *Part-Aware Multi-Scale Conv1D with Gated Multimodal Fusion and Head-Token Mixing Attention*. The main contributions are:

– A part-aware multimodal frame representation that combines MediaPipe landmarks of the left hand, right hand, body pose and selected face points with engineered geometry features and an explicit availability indicator for missing detections.

– Region-specific encoders that process each body part with residual depthwise temporal Conv1D blocks at two dilation rates, capturing short-range motion at more than one temporal scale.
– A learned gating mechanism that weights the hand, pose, face and geometry streams, conditioned on modality availability, before projection to a common 320-dimensional sequence.
– Temporal modelling with four Head-Token Mixing Attention (HTMA) blocks, and a sequence summary that combines a CLS-token representation with learned attention pooling.
– An evaluation on the 262-class INCLUDE benchmark using the official train/validation/test partition, reporting 98.04% Top-1 accuracy, 99.51% Top-5 accuracy, 98.09% macro-F1 and 98.19% balanced accuracy on the locked test set of 816 videos.

The rest of the paper is organised as follows. Section 2 reviews related work and states the research gap. Section 3 describes the INCLUDE dataset and the partition used. Section 4 presents the proposed method, and Section 5 the experimental setup. Section 6 reports and discusses the results, Section 7 outlines limitations and future work, and Section 8 concludes the paper.

2 Related Work

2.1 Sign Language Datasets

WLASL [Li et al. 2020] contains 21,083 web-collected videos of 2,000 American Sign Language (ASL) glosses performed by 119 signers, organised into subsets of 100, 300, 1,000 and 2,000 glosses. MS-ASL [Vaezi Joze and Koller 2019] provides a further 1,000-class ASL benchmark. RWTH-PHOENIX-Weather-2014T [Camgöz et al. 2018, Camgöz et al. 2020] contains continuous German Sign Language (DGS) weather broadcasts from nine signers, with a 1,066-sign vocabulary and German sentence translations. For ISL, earlier datasets were small, recorded under constrained conditions such as uniform backgrounds or cropped hands, and not publicly available [Sridhar et al. 2020]. INCLUDE [Sridhar et al. 2020] addressed this gap; it is described in Section 3.

2.2 Appearance-Based Recognition

Li et al. (2020) benchmarked two appearance-based models on WLASL: an ImageNet-pretrained VGG16 followed by a gated recurrent unit (GRU), and an I3D network [Carreira and Zisserman 2017] pretrained on ImageNet and Kinetics and fine-tuned on sign data. I3D was the stronger of the two, with 65.89% Top-1 accuracy on WLASL100 and 32.48% on WLASL2000. The INCLUDE baselines [Sridhar et al. 2020] use ImageNet-pretrained CNNs, including MobileNetV2 [Sandler et al. 2018], as frozen frame encoders applied to pose videos rendered by OpenPose, followed by a bidirectional LSTM (BiLSTM) decoder. For continuous recognition, Camgöz et al. (2020) embedded RGB frames with pretrained CNNs.

2.3 Keypoint-Based Recognition

Li et al. (2020) also evaluated a Pose-GRU, which concatenates the 2D coordinates of 55 OpenPose keypoints (13 upper-body joints and 21 per hand), and a pose-based temporal graph convolutional network (Pose-TGCN), which averages its features over time; they reached 55.43% Top-1 accuracy on WLASL100 and 23.65% on WLASL2000 and noted that pose-estimation errors may degrade recognition. SPOTER [Boháček and Hrúz 2022] normalises 54 two-dimensional body and hand landmarks, including five head points, extracted with Apple's Vision API; each hand is normalised in its own bounding box, missing landmarks are filled with zeros, and a Transformer classifies the 108-dimensional frame vectors. It reported 63.18% and 43.78% Top-1 macro accuracy on WLASL100 and WLASL300 and 100% on the Argentinian LSA64 dataset. The INCLUDE study [Sridhar et al. 2020] extracted 48 OpenPose keypoints per frame from the shoulders, arms, hands and fingers, discarding facial and leg keypoints; an optional Lucas–Kanade optical-flow imputation of missing keypoints did not improve accuracy, and an XGBoost classifier on the keypoint vector reached 89.1% on INCLUDE-50. Khartheesvar et al. (2024) used MediaPipe Holistic landmarks with an LSTM and reported 94.8% on INCLUDE-50 and 87.4% on INCLUDE.

2.4 Transformers and Self-Supervised Pre-Training

The Transformer [Vaswani et al. 2017] replaces recurrence with self-attention, allowing every position of a sequence to attend directly to every other position. SPOTER uses a Transformer encoder–decoder in which a single learned class query is decoded into the sign class [Boháček and Hrúz 2022]. SignBERT [Hu et al. 2021] treats the 2D pose of each hand in each frame as a visual token, embedded by a graph convolutional network together with temporal and hand-chirality embeddings. It pre-trains a Transformer encoder with BERT-style [Devlin et al. 2019] masked joint and masked frame modelling on the training data of its four benchmark datasets, reconstructing the masked poses through a decoder built on the MANO hand model [Romero et al. 2017], and then fine-tunes the encoder; its prediction head weights frame features with a simple attention mechanism before summing them. With hand pose alone it reported 39.40% per-instance Top-1 accuracy on WLASL2000. Camgöz et al. (2020) trained a Transformer encoder with a connectionist temporal classification (CTC) loss for continuous gloss recognition, jointly with an autoregressive decoder that generates German sentences, reporting a test word error rate of 24.49 and a BLEU-4 score of 21.80 on PHOENIX14T.

2.5 Combining Modalities and Body Regions

SignBERT combines its hand-pose model with a full-RGB or full-keypoint model by summing their prediction scores; with RGB this raised per-instance Top-1 accuracy on WLASL2000 to 54.69% [Hu et al. 2021]. Li et al. (2020) evaluated appearance and pose models side by side without combining them. In SPOTER [Boháček and Hrúz 2022] and the Pose-GRU of Li et al. (2020), all body parts share one concatenated frame vector rather than region-specific encoders.

2.6 Research Gap and Critical Comparison

The reviewed works leave several aspects of isolated ISL recognition open, although no single work exhibits all of them. First, keypoint representations are often limited to particular regions: the INCLUDE baselines discard facial keypoints [Sridhar et al. 2020], SignBERT models hand pose [Hu et al. 2021], and SPOTER uses body and hand landmarks with five head points [Boháček and Hrúz 2022]. Second, where several regions are used, they are typically concatenated into one frame vector [Boháček and Hrúz 2022, Li et al. 2020] rather than encoded separately. Third, the representations reported in [Sridhar et al. 2020, Li et al. 2020, Boháček and Hrúz 2022, Hu et al. 2021] are built from landmark coordinates; explicit geometric and motion descriptors such as bone lengths, joint angles, velocity and acceleration are not part of them. Fourth, missing detections are handled by zero-filling without an availability signal [Boháček and Hrúz 2022] or by imputation [Sridhar et al. 2020]. Fifth, temporal aggregation ranges from average pooling [Li et al. 2020] and a single class query [Boháček and Hrúz 2022] to attention-weighted summation [Hu et al. 2021]. Finally, among these works only the INCLUDE-based studies evaluate ISL [Sridhar et al. 2020, Khartheesvar et al. 2024].

The model proposed here addresses these aspects together: part-specific temporal encoders for the hands, pose and face; an explicit geometry stream; an availability-conditioned gated fusion; HTMA temporal modelling; and a summary that combines a CLS token with learned attention pooling, evaluated on the 262-class INCLUDE benchmark. Table 1 summarises the reviewed approaches along the dimensions that matter for comparison.

Work	Dataset (language), classes	Modality / spatial representation	Temporal model / aggregation	External pre-training	Protocol / metric	Reported result	Task
Sridhar et al. 2020	INCLUDE (ISL), 263; INCLUDE-50	48 OpenPose keypoints (face, legs discarded); pose-video frames	BiLSTM decoder; XGBoost on flattened keypoints	OpenPose; ImageNet MobileNetV2 (frozen)	Random stratified 20% test; accuracy	85.6% (INCLUDE); 94.5% (INCLUDE-50)	Isolated
Li et al. 2020	WLASL (ASL), 100–2,000	RGB (I3D, VGG16) or 55 OpenPose keypoints	3D conv.; GRU; TGCN with temporal average pooling	I3D: ImageNet + Kinetics; OpenPose	4:1:1 split; Top-1/5/10	I3D Top-1: 65.89% (100), 32.48% (2000)	Isolated
Boháček and Hrúz 2022	WLASL (ASL), 100/300; LSA64 (Argentinian), 64	54 2D body, head and hand landmarks; signing-space normalisation	Transformer encoder–decoder; class query	Apple Vision API; network randomly initialised	WLASL public split; LSA64 80/20; Top-1 macro	63.18%; 43.78%; 100%	Isolated
Hu et al. 2021	NMFs-CSL, SLR500 (Chinese); MSASL, WLASL (ASL)	2D hand pose of both hands (+ RGB or full pose, late fusion)	Transformer encoder; attention-weighted sum	MMPose; self-supervised pre-training	Per-instance and per-class Top-k	WLASL2000: 39.40% (hands); 54.69% (+ RGB)	Isolated
Camgöz et al. 2020	PHOENIX14T (DGS), 1,066 signs	RGB frames; CNN embeddings	Transformer encoder (CTC) + decoder	CNN features pre-trained on sign data	Dev / test; WER, BLEU	Test WER 24.49; BLEU-4 21.80	Continuous + translation
Khartheesvar et al. 2024	INCLUDE (ISL); INCLUDE-50	MediaPipe Holistic landmarks	LSTM	MediaPipe	Accuracy; macro-F1	87.4% (INCLUDE); 94.8% (INCLUDE-50)	Isolated

Work	Dataset (language), classes	Modality / spatial representation	Temporal model / aggregation	External pre-training	Protocol / metric	Reported result	Task
This work	**INCLUDE (ISL), 262**	**Hands 2 × 63, pose 132, face 276; geometry 469; availability 4**	**Part-aware Conv1D + 4 × HTMA; CLS + attention pooling**	**MediaPipe; recognition network trained from scratch**	**Official partition 3,121 / 347 / 816; locked test; Top-1/5, macro-F1, BAcc**	**98.04% Top-1; 99.51% Top-5**	**Isolated**

Table 1: Comparison of related work and the proposed model. Results are reproduced from each paper and use different datasets, vocabularies, data splits and metrics; they are not directly comparable across rows.

3 The INCLUDE Dataset

INCLUDE [Sridhar et al. 2020] was recorded at a school for the deaf with the help of seven senior students who were experienced ISL signers. The dataset paper reports 4,287 videos of 263 word signs grouped into 15 categories, such as adjectives, animals, colours, days and time, greetings, occupations and places. Each video shows a single isolated sign, beginning at the start of the sign and ending when it is completed. Videos were recorded at 1920 × 1080 resolution and 25 frames per second in a classroom with natural lighting, cluttered backgrounds and no constraints on clothing or signing style; the average video lasts 2.57 s [Sridhar et al. 2020].

Our experiments use the official INCLUDE train/validation/test partition. The class *secondnumber* is excluded, which reduces the label space from 263 to 262 classes. The resulting partition contains 3,121 training, 347 validation and 816 test videos, 4,284 videos in total, and the locked test set contains all 262 classes (Table 2). The test set was used only for the final evaluation.

Characteristic	Value
Videos / signs / categories [Sridhar et al. 2020]	4,287 / 263 / 15
Signers [Sridhar et al. 2020]	7 experienced ISL signers
Resolution and frame rate [Sridhar et al. 2020]	1920 × 1080, 25 fps
Average video length [Sridhar et al. 2020]	2.57 s
Excluded class	secondnumber
Classes in our experiments	262
Training / validation / test videos	3,121 / 347 / 816
Total videos used	4,284
Classes in the locked test set	262 of 262

Table 2: INCLUDE statistics and the partition used in this work.

4 Proposed Method

4.1 Overview

Figure 1 shows the complete pipeline. Each INCLUDE video is uniformly sampled into 64 frames, and MediaPipe extracts landmarks from every frame. Three per-frame inputs are formed: a 534-dimensional landmark vector, a 469-dimensional geometry

vector and a four-dimensional availability vector; for a mini-batch of B videos their shapes are (B, 64, 534), (B, 64, 469) and (B, 64, 4). Part-aware encoders process the four landmark regions and the geometry stream, a learned gate fuses the encoded streams with the availability representation, and the fused 400-dimensional frame features are projected to $d = 320$. A learned CLS token and learned positional embeddings are added, four HTMA blocks model the sequence, and the CLS representation and a learned attention-pooled summary are concatenated, projected from 640 to 320 dimensions and classified into one of 262 signs.

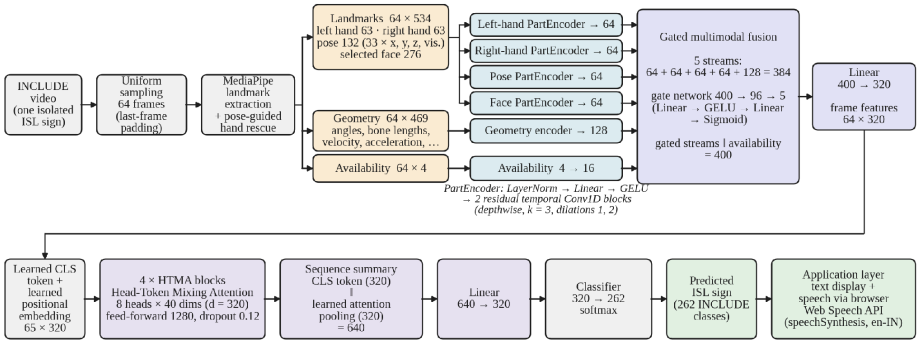


Figure 1: Architecture of the proposed model (IncludePartAwareConvHTMAV2). Top: 64-frame sampling, MediaPipe landmark extraction, the landmark, geometry and availability inputs, part-aware temporal Conv1D encoders and gated multimodal fusion (400 → 320). Bottom: CLS token and positional embedding, four HTMA blocks, CLS plus learned attention pooling (640 → 320), the 262-class classifier, and the application layer that displays the predicted sign and speaks it through the browser Web Speech API.

4.2 Temporal Sampling

Let a video contain F usable decoded frames. Frame indices are sampled uniformly across the complete video with linspace-style sampling:

$$i_k = (0,\ F - 1,\ T)_k,\quad k = 1,\ \ldots,\ T,\quad T = 64 \tag{1}$$

If the decoded sequence has fewer usable frames than required, the last successfully decoded frame is repeated so that every video is represented by exactly 64 frames. Each video is therefore uniformly sampled into a fixed sequence of 64 frames, with last-frame padding when necessary.

4.3 Landmark Representation

For every sampled frame t, MediaPipe [Lugaresi et al. 2019] landmarks are extracted for the left hand, the right hand, the body pose and a selected subset of face points. Each hand contributes 21 landmarks with (x, y, z) coordinates [Zhang et al. 2020], and

the pose contributes 33 landmarks with (x, y, z, visibility) [Bazarevsky et al. 2020]. The frame landmark vector is

$$x_t = \left[x_t^{LH};\ x_t^{RH};\ x_t^{P};\ x_t^{F} \right] \in \mathbb{R}^{534} \tag{2}$$

with 63, 63, 132 and 276 dimensions for the left hand, right hand, pose and selected face points respectively (Table 3). Missing landmark values are represented by zero-filled coordinates. Where possible, a pose-guided hand-rescue step recovers hand landmarks that the hand detector misses. A four-dimensional availability vector $\mathbf{a}_t \in \{0, 1\}^4$ indicates whether the relevant modalities are available in frame t, so that the model can distinguish a genuinely zero coordinate from a missing detection.

4.4 Geometry Features

A separate 469-dimensional geometry vector $\mathbf{g}_t$ encodes engineered spatial and temporal information derived from the landmarks: hand-centric coordinates, bone lengths, joint angles, spatial relationships, hand velocity and hand acceleration. The velocity and acceleration terms make short-range motion explicit instead of leaving it to be inferred from raw coordinates.

Stream	Input dimension	Encoded dimension
Left hand (21 × 3)	63	64
Right hand (21 × 3)	63	64
Pose (33 × 4)	132	64
Selected face	276	64
Geometry	469	128
Availability	4	16
Fused frame feature	384 + 16 = 400	320 (d)

Table 3: Per-frame inputs and encoder output dimensions.

4.5 Part-Aware Multi-Scale Temporal Encoding

The landmark vector is split into its four regions, and each region $p \in \{\text{LH, RH, P, F}\}$ is processed by its own PartEncoder over the 64-frame sequence X^p:

$$H^p = \left(\left(\left(W_p \left(X^p \right) \right) \right) \right) \tag{3}$$

where LN is layer normalisation [Ba et al. 2016], W_p is a linear projection, GELU is the Gaussian error linear unit [Hendrycks and Gimpel 2016], and each region is mapped to 64 dimensions per frame. Each residual temporal block (RTB) applies

$$(Z) \ = Z + (((((((Z)\ )\ )\ )\ )\ )\ ) \tag{4}$$

where $\mathrm{DW}_{k,\delta}$ is a depthwise Conv1D along time with kernel size $k = 3$ and dilation δ, PW_{exp} and PW_{proj} are pointwise Conv1D layers that expand and then project the channels, and Drop is dropout. Depthwise followed by pointwise convolution keeps the temporal filters inexpensive [Sandler et al. 2018]. The two blocks use dilations of 1 and 2, so the encoder captures motion over more than one temporal scale. The

geometry stream is encoded to 128 dimensions per frame by a geometry branch, and the availability vector is projected from 4 to 16 dimensions.

4.6 Gated Multimodal Fusion

The five main encoded streams, $\mathbf{h}_t^{\mathrm{LH}}$, $\mathbf{h}_t^{\mathrm{RH}}$, $\mathbf{h}_t^{\mathrm{P}}$, $\mathbf{h}_t^{\mathrm{F}}$ (64 dimensions each) and $\mathbf{h}_t^{\mathrm{G}}$ (128 dimensions), total 384 dimensions per frame. Together with the 16-dimensional availability representation $\mathbf{u}_t$, they form a 400-dimensional input to a gating network that produces five gates:

$$\gamma_t = \sigma\left(W_2\left(W_1\left[h_t^{LH};\ \ldots;\ h_t^{G};\ u_t\right]\right)\right) \in (0,\ 1)^5 \tag{5}$$

where W_1 maps 400 to 96 dimensions, W_2 maps 96 to 5, and σ is the sigmoid function. Each gate scales its stream, and the gated streams are combined with the availability representation and projected to the model dimension:

$$z_t = W_f\left[\gamma_{t,1}h_t^{LH};\ \ldots;\ \gamma_{t,5}h_t^{G};\ u_t\right],\quad z_t \in \mathbb{R}^{320} \tag{6}$$

with W_f mapping the 400-dimensional fused representation to d = 320. The gates allow the model to reduce the influence of a stream that is unreliable, for example when a hand is missing, while the availability representation informs the gate of which modalities were detected.

4.7 Head-Token Mixing Attention

A learned CLS token [Devlin et al. 2019] $\mathbf{c}$ is prepended to the 64 fused frame features, and learned positional embeddings E are added to give the input sequence

$$Z_0 = \left[c;\ z_1;\ \ldots;\ z_{64}\right] + E,\quad Z_0 \in \mathbb{R}^{65\times 320} \tag{7}$$

Temporal modelling is performed by four HTMA blocks, the temporal self-attention mechanism of the final model. Each block uses eight attention heads of dimension 40 (320 / 8 = 40), a feed-forward dimension of 1,280 and dropout of 0.12. Within head j, attention between the 65 tokens follows the scaled dot-product form [Vaswani et al. 2017]

$$A_j = \left(\frac{Q_j K_j^{\mathrm{T}}}{\sqrt{40}}\right)V_j,\quad j = 1,\ \ldots,\ 8 \tag{8}$$

HTMA extends this multi-head attention with a head-token mixing operation. Because every frame token can attend to every other token, the block relates the start, middle and end of a sign directly; this matters when two signs share a handshape and differ mainly in their motion.

4.8 Sequence Summary and Classification

After the four HTMA blocks, two 320-dimensional summaries are formed: the output of the CLS token, $\mathbf{h}_{\mathrm{CLS}}$, and a learned attention pooling over the 64 temporal tokens $\mathbf{h}_1, \ldots, \mathbf{h}_{64}$:

$$\alpha_t = \left(s\left(h_t\right)\right), \quad p = \sum_t \alpha_t h_t \tag{9}$$

where $s(\cdot)$ is a learned scoring function. The two summaries are concatenated (320 + 320 = 640), projected to 320 dimensions and classified:

$$\hat{y} = \left(W_c W_r\left[h_{CLS};\ p\right] + b_c\right) \tag{10}$$

where W_r maps 640 to 320 dimensions and W_c maps 320 to the C = 262 INCLUDE classes. Temporal aggregation is therefore neither global average pooling nor last-step pooling, but a combination of the CLS representation and learned attention pooling.

4.9 Training Objective

The model is trained with a class-weighted cross-entropy loss with label smoothing [Szegedy et al. 2016]. With class weights w_c and smoothing factor $\varepsilon = 0.10$, the smoothed target for a sample of class y is $q_c = (1 - \varepsilon)\ \mathbb{1}[c = y] + \varepsilon / C$, and

$$\mathcal{L} = -\sum_{c=1}^{C} w_c q_c \log log\ \hat{y}_c \tag{11}$$

4.10 Text, Speech and Real-Time Application

The recognition model outputs a class index, which the application maps to its sign label and displays as text. Speech is produced in the browser through the Web Speech API (speechSynthesis) with a preference for an en-IN voice. This component is part of the application layer: it adds no learnable parameters, and the reported 98.04% is a recognition accuracy, not a measure of speech generation. The project also includes a real-time webcam application that applies the recognition pipeline to live video; it is a deployment component, and no webcam accuracy is reported in this paper.

5 Experimental Setup

5.1 Training Configuration

Table 4 lists the training configuration. The model is optimised with AdamW [Loshchilov and Hutter 2019] at a learning rate of 2.5×10^{-4} and weight decay of 2×10^{-4}, with a five-epoch warm-up followed by cosine decay [Loshchilov and Hutter 2017] to a minimum learning rate of 1×10^{-6}. Training runs for at most 70 epochs with a batch size of 32, gradient clipping at 1.0, an exponential moving average (EMA) of the weights with decay 0.995, and early stopping with a patience of 15 epochs. The random seed is 42. The checkpoint with the best validation performance, obtained at epoch 38 and saved as include262_conv1d_htma_v2_seed42_best.pt, was selected; epoch 38 is the selected best checkpoint, not the final training epoch. The locked test set was then used once for the final evaluation.

Setting	Value
Model	IncludePartAwareConvHTMAV2

Setting	Value
Frames per video	64 (uniform, last-frame padding)
Model dimension d	320
HTMA blocks / heads / head dim.	4 / 8 / 40
Feed-forward dim. / dropout	1,280 / 0.12
Temporal Conv1D kernel / dilations	3 / 1 and 2
Optimiser	AdamW
Learning rate / min. learning rate	2.5×10^{-4} / 1×10^{-6}
Weight decay	2×10^{-4}
Schedule	5-epoch warm-up, cosine decay
Batch size / max. epochs	32 / 70
Gradient clipping / EMA decay	1.0 / 0.995
Loss	Weighted cross-entropy, label smoothing 0.10
Early stopping patience	15 epochs
Random seed	42
Selected checkpoint	Epoch 38 (best validation)

Table 4: Model and training configuration.

5.2 Evaluation Metrics

Let N be the number of test videos, y_i the true class of video i, and $\hat{y}_i$ the predicted class scores. Top-k accuracy is the fraction of videos whose true class is among the k highest-scoring classes:

$$Acc@k = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\left[y_i \in top - k\left(\hat{y}_i\right)\right] \tag{12}$$

We report $k = 1$ and $k = 5$. Because INCLUDE has 262 classes with a few test videos each, overall accuracy could hide weak classes, so we also report macro-F1, the unweighted mean of per-class F1 scores, and balanced accuracy, the unweighted mean of per-class recall:

$$Macro - F1 = \frac{1}{C}\sum_{c} F1_c, \quad BAcc = \frac{1}{C}\sum_{c} Recall_c \tag{13}$$

Both class-balanced metrics are averaged over the classes present in the test set; all 262 classes are present in the locked test set.

6 Results and Discussion

6.1 Recognition Performance

Table 5 reports the final results on the locked INCLUDE test set. The model classified 800 of the 816 test videos correctly at Top-1, an accuracy of 98.04%, with 16 misclassified videos. The true sign was among the five highest-scoring classes for 812 videos, a Top-5 accuracy of 99.51%; 12 of the 16 Top-1 errors therefore still ranked the correct sign within the top five. Macro-F1 (98.09%) and balanced accuracy (98.19%) are close to Top-1 accuracy, which indicates that performance is spread

across classes rather than concentrated in a few of them. The complete evaluation of the 816 test videos took 6.60 s.

Metric	Value
Top-1 accuracy	**98.04%** (800 / 816)
Top-5 accuracy	99.51% (812 / 816)
Macro-F1	98.09%
Balanced accuracy	98.19%
Test videos / classes	816 / 262
Top-1 errors	16
Total evaluation time	6.60 s
Checkpoint	Epoch 38 (best validation)

Table 5: Final results on the locked INCLUDE test set (262 classes).

6.2 Comparison with Results Reported on INCLUDE

Table 6 places our result alongside accuracies previously reported on the full INCLUDE dataset. The INCLUDE study reports 85.6% for its best pipeline, which combines OpenPose-rendered pose videos, a frozen MobileNetV2 encoder, a BiLSTM decoder and data augmentation [Sridhar et al. 2020], and Khartheesvar et al. (2024) report 87.4% with MediaPipe Holistic landmarks and an LSTM.

These figures are reference points rather than a controlled comparison. The INCLUDE paper evaluates 263 classes and describes a random stratified 20% test subset [Sridhar et al. 2020], whereas our evaluation uses 262 classes and the official train/validation/test partition with a locked test set of 816 videos; whether the INCLUDE paper's test set coincides with this partition is not established here. Representations, preprocessing, augmentation and training protocols also differ. A strict claim of improvement would require re-running the methods under an identical partition and metric. Within these limits, the results suggest that part-aware landmark encoding, explicit geometry, availability-aware gated fusion and HTMA temporal modelling form an effective design for large-vocabulary isolated ISL recognition.

Method	Representation / model	Classes	Acc. (%)
Sridhar et al. 2020	Pose points + XGBoost (no augmentation)	263	63.1
Sridhar et al. 2020	OpenPose pose video + MobileNetV2 + BiLSTM (augmented)	263	85.6
Khartheesvar et al. 2024	MediaPipe Holistic + LSTM	Full set	87.4
This work	**Part-aware Conv1D + gated fusion + HTMA**	**262**	**98.04**

Table 6: Accuracies reported on the full INCLUDE dataset. Partitions, class counts and preprocessing differ between rows; see Section 6.2.

6.3 Comparative Performance Analysis

Figure 2 summarises published models that report two or more performance metrics, together with our model. Where a paper reports Top-1 accuracy, Top-5 accuracy or macro-F1, the values are plotted as reported; metrics a paper does not report are

marked n/r. The WLASL2000 baselines of Li et al. (2020) (VGG-GRU, Pose-GRU, Pose-TGCN and I3D) reach between 8.4% and 32.5% Top-1 accuracy, and SignBERT [Hu et al. 2021] reaches 39.4% Top-1 with hand pose alone and 54.7% when fused with RGB, illustrating how difficult large-vocabulary isolated recognition remains. On INCLUDE, the MediaPipe + LSTM model of Khartheesvar et al. (2024) reports 87.4% Top-1 accuracy and 86.6% macro-F1, and 94.8% and 93.5% on INCLUDE-50. Sridhar et al. (2020) report accuracy only (85.6% / 94.5%) and are therefore not plotted.

Our model is the only one in this group that reports all three metrics, with 98.04% Top-1 accuracy, 99.51% Top-5 accuracy and 98.09% macro-F1, and it also reaches 98.19% balanced accuracy on 262 classes. Because benchmarks, vocabulary sizes and evaluation splits differ between these works, Figure 2 summarises published results rather than a single controlled experiment, and it should be read together with the caveats in Section 6.2.

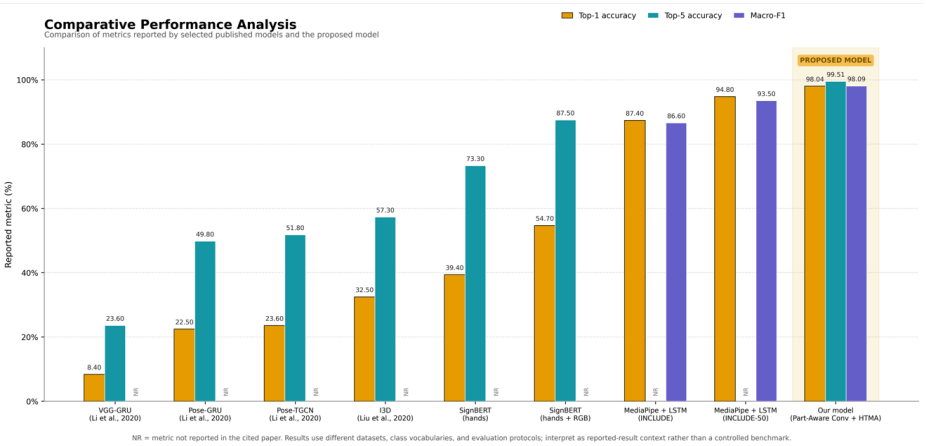


Figure 2: Comparative performance analysis of the proposed model against published models that report two or more performance metrics (Top-1 accuracy, Top-5 accuracy and macro-F1). n/r = metric not reported by that paper. SignBERT figures are per-instance accuracy. Our result: locked test set, 816 videos, 262 classes, epoch-38 EMA checkpoint.

6.4 Relation to Work on Other Sign Languages

SPOTER, SignBERT, the WLASL baselines and Sign Language Transformers were evaluated on American, Argentinian, Chinese and German sign language datasets with different vocabularies and, for Camgöz et al. (2020), a continuous-recognition task measured by word error rate and BLEU [Boháček and Hrúz 2022, Hu et al. 2021, Li et al. 2020, Camgöz et al. 2020]. Their numerical results are therefore not directly comparable with ours. The architectural comparison in Table 1 is more informative: the proposed model shares attention-based temporal modelling with SPOTER, SignBERT and Sign Language Transformers, and learned temporal weighting with SignBERT's prediction head, but differs in encoding each body region separately,

adding explicit geometry and availability streams, and fusing the streams through learned gates before HTMA.

6.5 Discussion

Each component targets a specific difficulty of isolated ISL recognition. Separate part encoders let the hands, pose and face be modelled at their own granularity, which suits a largely two-handed language with meaningful facial cues. The geometry stream supplies relationships that are otherwise implicit in raw coordinates, such as joint angles and hand velocity. The availability vector and gates give the model a way to discount streams that are missing in some frames. HTMA and the combined CLS and attention-pooled summary provide flexible temporal aggregation over the full 64-frame sequence. The results in Table 5 are consistent with this design, but the paper does not include an ablation study, so the individual contribution of each component has not been measured.

7 Limitations and Future Work

The evaluation has several limitations. The primary benchmark is isolated sign recognition over the 262-class INCLUDE label space, so the results do not by themselves establish continuous, sentence-level recognition or translation [Camgöz et al. 2020]. All experiments use a single dataset recorded by seven signers [Sridhar et al. 2020], and signer-independent generalisation has not been tested separately. The model depends on MediaPipe landmark extraction, so occlusion, extreme viewpoints, poor lighting and landmark-detection failures may reduce performance. Real-world webcam use involves a domain shift from the INCLUDE recording conditions and is a separate deployment challenge that this benchmark does not measure. Finally, the contribution of each component has not been isolated by ablation.

Future work will address these points through ablation studies of the part encoders, geometry stream, gating and HTMA; signer-independent evaluation; analysis of the 16 misclassified test videos; a documented evaluation of the real-time webcam application; and extension towards continuous ISL recognition.

8 Conclusions

This paper presented a landmark-based model for isolated Indian Sign Language recognition that combines part-aware multi-scale temporal Conv1D encoders for the hands, pose and face, an explicit geometry stream, availability-aware gated multimodal fusion, four Head-Token Mixing Attention blocks, and a sequence summary that joins a CLS token with learned attention pooling. On the locked test set of the official INCLUDE partition, with 816 videos and 262 classes, the model achieved 98.04% Top-1 accuracy, 99.51% Top-5 accuracy, 98.09% macro-F1 and 98.19% balanced accuracy. Combined with text display and browser-based speech output, the system moves isolated ISL recognition a step closer to a practical communication aid.

Acknowledgements

The author thanks the creators of the INCLUDE dataset and the students of the school for the deaf who recorded it for making their work publicly available.

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